

Fuel for living

All animals need energy to live and they get it by eating food. Many feed on leaves, fruit, and seeds. Others eat other animals or dead remains. Some animals can survive for weeks between meals, but most need to find food every day. Food energy is processed by living things in a food chain, which begins when plants make food from sunlight. This energy is passed on when animals feed on the plants.

Wing muscles powered
by food energy

Feeding tools

Meat-eaters and plant-eaters have varied equipment for gathering and breaking down their food.

Mammals, for instance, have differently shaped teeth depending on their diet. Lions have pointed canine teeth that grip their prey, and scissorlike carnassial teeth that slice up meat. In contrast, rabbits have continually growing incisor teeth that gnaw grass and rows of molar teeth that grind up plant matter at the backs of their jaws.

Lion skull

Carnassial tooth has
jagged, slicing edge

Canine tooth has
sharp point

Sharp incisor tooth
has flat, snipping edge

Flat molar tooth has
raised, grinding ridges

Rabbit skull

Berry banquet

Fieldfares feast on berries during winter months, when other food can be hard to find.

This harvest is created from sunlight, which trees collect with their leaves through a process called photosynthesis. Like all plants, trees use this solar energy to grow and to make energy-rich substances, including sugars and starch. Fieldfares and other fruit-eaters eat this food, and, in return, they help to scatter a tree's seeds.

Food chain



Solar energy

The Sun supplies energy to most living things. Plants use sunlight to make food, which is then passed along the food chain as chemical energy when animals eat the plants or each other.



Grass

In the vast prairies of North America, sunlight is collected by grasses and other plants. Grasses use solar energy to grow leaves and seeds, as well as a dense network of roots.



Grasshopper

The prairies teem with grasshoppers and other plant-eating insects. Their numbers explode in summer when plants are growing fast, creating a huge supply of food.



Leopard frog

At night, the leopard frog emerges from pools to hunt. Its diet includes grasshoppers, as well as many other small prairie animals, from earthworms to centipedes.



Garter snake

This slender snake lives in a wide variety of habitats, particularly in areas where it is damp. The garter snake hunts and catches leopard frogs on land or swims after them in water.



Harrier

Sweeping low over the prairies, the northern harrier snatches up snakes in its talons. This hawk is a top predator, meaning that it has no natural enemies.